INCH-POUND

MIL-PRF-1/1727B
w/AMENDMENT 1
5 October 2022
SUPERSEDING
MIL-PRF-1/1727B
6 January 2015

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, MICROWAVE, NEGATIVE GRID

TYPE DOD-018 *

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: Triode, planar.
See figure 1.
Mounting position: Any.
Weight: 1 ounce nominal.

ABSOLUTE RATINGS:

Parameter:	Ef	F1	Eb	Ec	Ehk	ib	Ib	Ic	ik	ic	Pp	Pi	tp	Du	tk	T (seal)	Barometric pressure, reduced
Unit:	V	GHz	V dc	V dc	V dc	a	mA dc	mA dc	a	a	W	W	µs	- - -	sec	1/ °C	2/ mmHg
Osc or amp 3/																	
Maximum:	4/ 5.7	2.5	350	-50	±90	---	35	15	---	---	8.0	12	---	---	---	175	20
Minimum:	4.5	---	---	---	---	---	---	---	---	---	---	---	---	---	30	---	---
Pulse modulated, Osc or amp 5/																	
Maximum:	4/ 5.7	2.5	275	-50	±90	---	35	15	---	---	6.0	9.5	---	---	---	175	20
Minimum:	4.5	---	---	---	---	---	---	---	---	---	---	---	---	---	30	---	---
Anode pulsed, Osc or amp 6/																	
Maximum:	4/ 6.3	3.5	---	-100	±90	7/ 2.5	2.5	1.25	3.75	1.25	7.5	7.5	2.0	0.001	---	175	250
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---	---	---	---	60	---	---
Class A, RF amp																	
Maximum:	6.6	---	350	-25	±90	---	35	---	---	---	8.0	12	---	---	---	175	20
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---	---	---	---	0	---	---
Test conditions:	6.3	---	350	Adj	---	---	35	---	---	---	---	---	---	---	120 min	9/ ---	---

See footnotes at end of table I.

GENERAL:

Qualification: Not Required.
Holding period: t = 72 hours.
This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.

* Replaces G. E. tube type Y-2052.

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

FSC 5960

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:26Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and inspection.

Inspection	MIL-STD-1311 method	Conditions	Symbol	Limits		Unit
				Min	Max	
Conformance inspection, part 1		10/				
Pulsed oscillation	---	F = 3.45 GHz (min); Epy = 3,000 V; Rg/Ib = 2.5 mA dc; tpv = 1.0 μs ±10 percent; trv = 0.1 μs (max); tfv = 0.2 μs (max); prr/Du = 0.001 ±5 percent; Ef = 6 V 12/ 13/	Po	1.90	---	W (useful)
Insulation of electrodes	1211	Ef = 6.3 V 14/	Rgp Rgk	250 25	--- ----	MegΩ MegΩ
Emission	1231	Eb = Ec/Is = 40 ma dc	Eb = Ec	---	4.0	V dc
Pulsing emission	1231	Prr = 500 ±10 percent; tp = 1 ±0.1 μs; eb = ec/is = 5 a	eb = ec	---	110	v
Electrode voltage (1) (grid)	1261		Ec	-1.90	-4.40	V dc
Total grid current	1266		Ic	---	0.5	μA dc
Conformance inspection, part 2						
Electrode voltage (2) (grid)	1261	Ec/Ib = 2 mA dc	Ec	-3.60	-8.50	V dc
Heater current	1301		If	840	960	mA
Transconductance	1306		Sm	21,000	30,000	μmhos
Resonance test	---	No voltages applied 8/ 11/	---	---	---	---
Amplification factor	1316		Mu	60	110	---
Direct-interelectrode capacitance	1331	No voltages applied; use shielded socket in accordance with Drawing 260-JAN.	Cgp Cgk Cpk	2.10 4.85 ---	2.4 5 5.75 0.045	pF pF pF
Heater-cathode leakage	1336	Eb = Ec = 0	Ihk	---	100	μA dc

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:26Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and inspection - Continued.

Inspection	MIL-STD-1311 method	Conditions	Symbol	Limits		Unit
				Min	Max	
Conformance inspection, part 3						
Life test	---	Group S; t = 1,000 hours; Ef = 6.3 V (no other voltages) 15/	---	---	---	---
Life-test end point:						
Pulsed oscillation	---		ΔPo	---	25	%
Interelectrode capacitance (gk)	---		ΔCgk	---	0.35	pF

1/ Sufficient conduction and convection cooling shall be provided to limit the envelope temperature to the specified maximum to +175°C under all operating conditions.

2/ Operation at this altitude is possible in a suitably designed circuit.

3/ Modulation essentially negative may be used if the positive peak of the audio frequency envelope does not exceed 115 percent of the carrier conditions.

4/ This tube operates at frequencies where transit-time effects of the electron current shall be considered. The principal effects influencing the tube operation are the decrease in power output and operating efficiency with increase in frequency, and the bombardment and heating of the cathode by electrons from the region of the grid, which can be sufficiently severe to result in short tube life and erratic operation. Operating frequency, circuit design and adjustment, grid bias, grid current, average power input and the operating voltage contribute to some degree to the magnitude of the resulting cathode bombardment. There is a heater voltage which will maintain the cathode at the correct operating temperature for a set of operating conditions. If the conditions of operation result in appreciable cathode back-heating, it may be necessary to initiate dynamic operation of the tube at normal heater voltage, followed by a reduction of heater voltage to the proper value. A maximum variation of ±5 percent in heater voltage is recommended where consistent operation and extended tube life is a factor.

5/ Carrier conditions for use with a maximum modulation factor of 1.0.

6/ Tube shall not be operated for more than 4 microseconds in any 2,000 microsecond interval. Tube shall not be grid-pulsed beyond the CW ratings.

7/ The regulation of series-anode-supply impedance, or both, shall limit the instantaneous peak current, with the tube considered as a short circuit, to a maximum of 10 times the specified maximum current rating.

8/ Other tube contact configurations may be used provided the tube contact area remains unchanged and the socket, jig, or cavity gives equal performance. Mounting of the socket, jig, or cavity may be at the option of the manufacturer.

9/ During any test the maximum temperature of any part of the envelope shall be limited to +175°C maximum by conduction or forced convection cooling.

10/ All tests listed under conformance inspection, part 1, are to be performed at the conclusion of the holding period.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:26Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and inspection - Continued.

11/ Grid-anode resonance: Test in cavity in accordance with DSCC drawing D64102. Cavity shall resonate at 1.74 GHz ±2.5 MHz with tuning slug in accordance with DSCC drawing B64103 at TA = +25° ±5°C.

Grid-cathode resonance: Test in cavity in accordance with DSCC drawing D64102. Cavity shall resonate at 1.793 GHz ±2.5 MHz with tuning slug in accordance with DSCC drawing B64103 at TA = +25° ±5°C.

When plotted on graphs of resonant frequency versus grid-anode capacitance and resonant frequency versus grid-cathode capacitance, the tube under test (TUT) shall be represented by a point within a parallelogram whose four corners are located by the following points:

Points	Capacitance (pF)		Frequency (GHz)	
	Cgp	Cgk	Fgp	Fgk
1	2.10	4.85	1.928	1.879
2	2.10	4.85	1.947	1.903
3	2.45	5.75	1.895	1.845
4	2.45	5.75	1.914	1.869

12/ Test in cavity in accordance with Drawing 253-JAN Measure useful power output with the coupling and tuning adjusted for maximum power output. The tube shall operate under the given conditions and after the required circuit adjustments for a minimum of 10 seconds without arcing or instability as evidenced by pronounced variations in the average anode current meter or the power output meter. If temporary arcing does occur during the 10-second period, the tube shall subsequently operate for 60 seconds without arcing. The test may be discontinued at the end of the 10-second period if the power output has not completely stabilized but is rising. If the power output is falling, the test will be continued until a stable reading is obtained. For qualification purposes, a stabilized reading shall be obtained. All readings shall be “nonsquegging” power.

13/ The pulse characteristics are defined as follows:

a. epy is the smooth peak voltage excluding spike (see MIL-STD-1311, method 1296).

b. The interval of trv and tfv is between 20 and 85 percent of the smooth peak value of the voltage pulse.

c. The definition of tp is applicable to tpv.

d. The parameter tpv, tfv, and trv may be measured with the tube replaced by a 1,000-ohm non-inductive resistor.

The load VSWR shall be 1.5 maximum.

14/ The insulation between the grid and the cathode shall be measured with a voltage of 100 V dc. The other insulation test shall be made in accordance with MIL-STD-1311, method 1211.

15/ The test shall be conducted under room ambient conditions. No cooling of the tube and no heat sink shall be permitted. The filament voltage shall be cycled “on” 1 hour 45 minutes and “off” 15 minutes. The specified life test duration applies to accumulated “on” time.

.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:26Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and inspection - Continued.

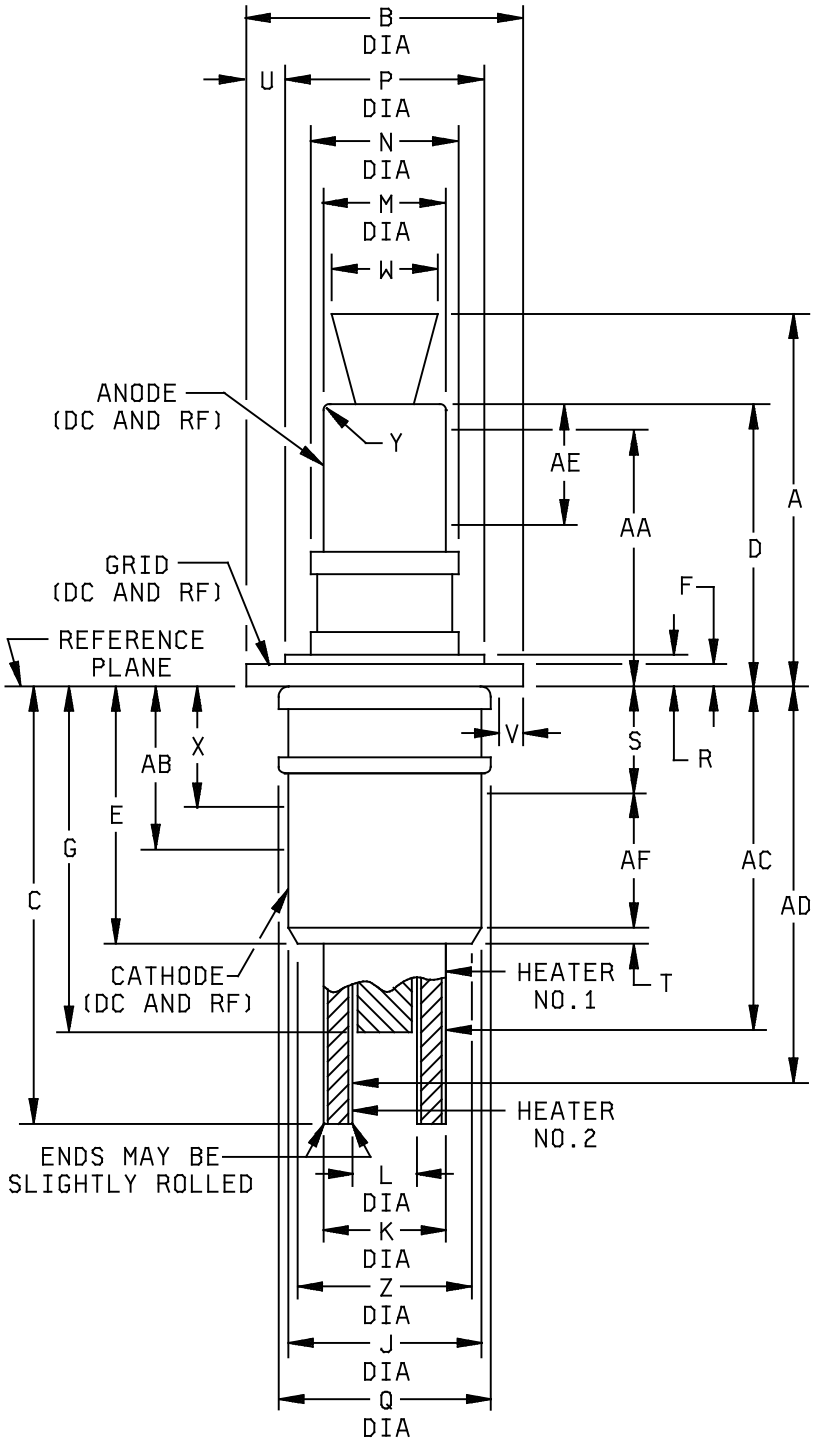


FIGURE 1. Outline drawing of electron tube type DOD-018.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:26Z
Check the source to verify that this is the current version before use.

Ltr	Dimensions				
	Inches		Millimeters		Notes
	Min	Max	Min	Max	
	conformance		inspection,	part 2	
A	---	.250	---	6.35	
B	.810	.818	20.57	20.77	
C	1.219	1.281	30.96	32.53	
D	.953	.984	24.21	24.99	
E	.750	.813	19.05	20.65	
F	.070	.078	1.78	1.98	3, 4
G	---	1.016	---	25.80	
J	.539	.549	13.69	13.94	5
K	.318	.328	8.08	8.33	
L	.180	.190	4.57	4.82	
M	.365	.371	9.27	9.42	
AE	.429	---	10.90	---	3
	Conformance	inspection,	part 3		
N	---	.453	---	11.50	
P	.560	.570	14.22	14.47	
Q	---	.609	---	15.46	
R	.077	.097	1.96	2.46	
S	---	.266	---	6.75	
T	.015	.090	0.38	2.28	
W	---	.285	---	7.24	1
X	---	.375	---	9.52	6
Z	.490	.520	12.45	13.20	
		Reference	dimensions		
U	0.094		2.39		3, 4
V	0.094		2.39		3, 4
Y	0.016 R		0.41 R		
AA	0.750		19.05		2
AB	0.547		13.89		2
AC	1.000		25.40		2
AD	1.109		28.17		2
AF	0.453		11.51		3

FIGURE 1. Outline drawing of electron tube type DOD-018 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:26Z
Check the source to verify that this is the current version before use.

NOTES:

1. Exhaust tubulation shall not be subjected to any mechanical stress.
2. Eccentricity shall be gauged at points designated and shall be as follows:

Contact	Eccentricity		Reference
	Inches	Millimeters	
Anode	.010 max	.25	Grid contact
Cathode	.010 max	.25	Grid contact
Anode-cathode	.020 TIR	.51	Grid contact
Heater No. 1	.015 max	.38	Grid contact
Heater No. 2	.015 max	.38	Grid contact
Heater No. 1 and 2	.030 TIR	.76	Grid contact

3. Contact surface shall be confined to this area.
4. Only these surfaces shall be used for tube stops or clamping.
5. Maximum diameter shall not be increased by solder.
6. Tube marking shall be confined to this area.

FIGURE 1. Outline drawing of electron tube type DOD-018 - Continued.

Referenced Documents. In addition to MIL-PRF-1, this specification sheet references:
MIL-STD-1311
Drawing 253-JAN
Drawing 260-JAN
DSCC drawing D64102
DSCC drawing B64103

NOTE: To obtain copies of JAN drawings, please send a request via email to TubesAmps@dla.mil.

Amendment notations. The margins of this specification sheet are marked with vertical line to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2022-021)

Review activities:
Navy - AS, CG, MC, OS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:26Z
Check the source to verify that this is the current version before use.